Assignment – 2.13

Troubleshooting Network Services

Q - 1. Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

- **DNS Resolution Failure:**

  Aim: To simulate DNS failure by modifying the hosts file.

- **Steps:**

  1. Open Notepad as Administrator.
  2. Open the hosts file:
     C:\Windows\System32\drivers\etc\hosts
  3. Add this line:

     192.0.2.1 www.spotify.com
  4. Save the file.
  5. Open CMD and type:

     ipconfig /flushdns

     ping www.spotify.com

- **Expected Output:**

  Pinging www.spotify.com [192.0.2.1]

  Request timed out.

  Request timed out.

- **Observation:**

  The domain points to the wrong IP address, so the ping fails.

- **Solution:**

  Remove the incorrect entry from the hosts file and flush DNS again.

- **Result:**

  Successfully simulated a DNS resolution failure.

Q - 2. Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

- **DHCP Server with Incorrect Default Gateway:**
- **Steps:**

1. Open Cisco Packet Tracer.
2. Add a router, switch, server, and PC.
3. Connect all devices.
4. Enable DHCP on the server.
5. Enter an incorrect default gateway (192.168.1.100).
6. Set the PC to DHCP mode.
7. Check the IP address and gateway using ipconfig.

- **Problem:**

The PC receives the wrong default gateway and cannot access other networks.

- **Solution:**

1. Open DHCP settings.
2. Change the gateway to the correct IP (192.168.1.1).

3. Renew the IP address.

4. Check the gateway again.

- **Result:**

  The wrong gateway was identified and corrected. The PC can now communicate with other networks.

Q - 3. Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

- **DNS Troubleshooting:**

  Problem: Websites open using IP addresses but not website names.

- **Steps:**

  1. Open Command Prompt.
  2. Type nslookup www.google.com.
  3. Check whether the DNS server is working.
  4. If DNS is not working, check the DNS IP address.
  5. Set the correct DNS address, such as 8.8.8.8.
  6. Type ipconfig /flushdns.
  7. Try opening the website again.

- **Solution:**

  Correct the DNS server IP address.

- **Result:**

  Websites open successfully using their names.

Q - 4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it. Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

- **DHCP Troubleshooting (APIPA Address):**
- **Possible Causes:**

1. DHCP server is OFF.
2. Network cable is disconnected.
3. No free IP addresses available.
4. Router or switch is not working.
5. DHCP server is unreachable.

- **Steps to Solve:**

1. Check the network cable connection.
2. Make sure the DHCP server is ON.
3. Check if free IP addresses are available.
4. Check the router and switch.
5. Run ipconfig /release.
6. Run ipconfig /renew.
7. Check whether the device gets a correct IP address.

Q - 5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

➢ Common DNS Issues:

No.	Issue	Symptoms	Causes	Solutions
1	DNS Server Down	Websites do not open.	Server not working.	Restart DNS server.
2	Wrong DNS Address	Websites cannot open by name.	Incorrect DNS IP.	Set correct DNS IP.
3	DNS Error	Website opens by IP only.	DNS problem.	Check DNS settings.

➢ Common DHCP Issues:

No.	Issue	Symptoms	Causes	Solutions
1	DHCP Server Down	Device gets no IP.	DHCP service stopped.	Restart DHCP service.
2	IP Address Exhausted	New devices get no IP.	All IPs are used.	Increase IP range.
3	Wrong Gateway	Internet does not work.	Incorrect gateway.	Set correct gateway.